

Unit 17

Business Process

Support

BTECH HND in Computing

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What is a business process?



A business process is a **collection** of related **events**, **activities**, and **decisions**, that involve several actors and resources, and that collectively lead to an outcome that is of value to an organization or its customers.

What is a business process?

- Business processes are what companies do whenever they deliver a service or a product to customers.
- A company can outperform another company offering similar kinds of service if it has better processes and executes them better.



Activity

Find out the business processes in

- ✓ IT company
- ✓ Bank
- ✓ e-commerce platform
- ✓ BCAS campus



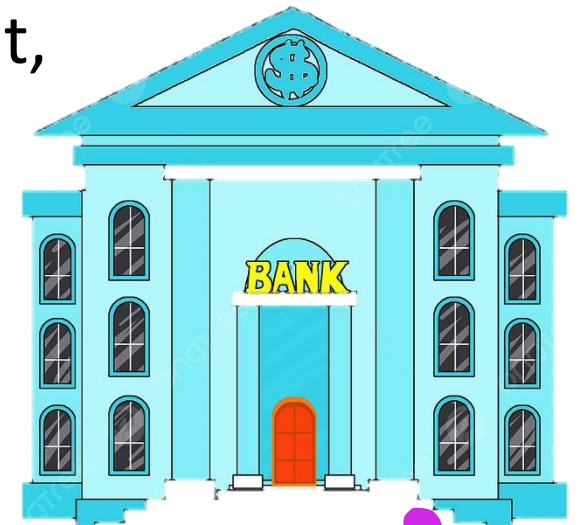
Business processes in an IT company

- Recruitment
- Project management
- Software development
- Software testing
- Knowledge management
- Security
- Client management
- Marketing



Business processes in a bank

- Open a new customer account
- Loan Approval Process
- Transaction process → cash withdrawal, deposit, money exchange, etc.
- Credit Card Issuance → credit card, debit card
- ATM management



Business processes in a school

- Admission process
- Attendance tracking
- Classroom instructions
- Parent-Teacher Conferences
- Student Support Services → Counseling, special education services, health services, and extracurricular activities



Business processes in an e-commerce platform

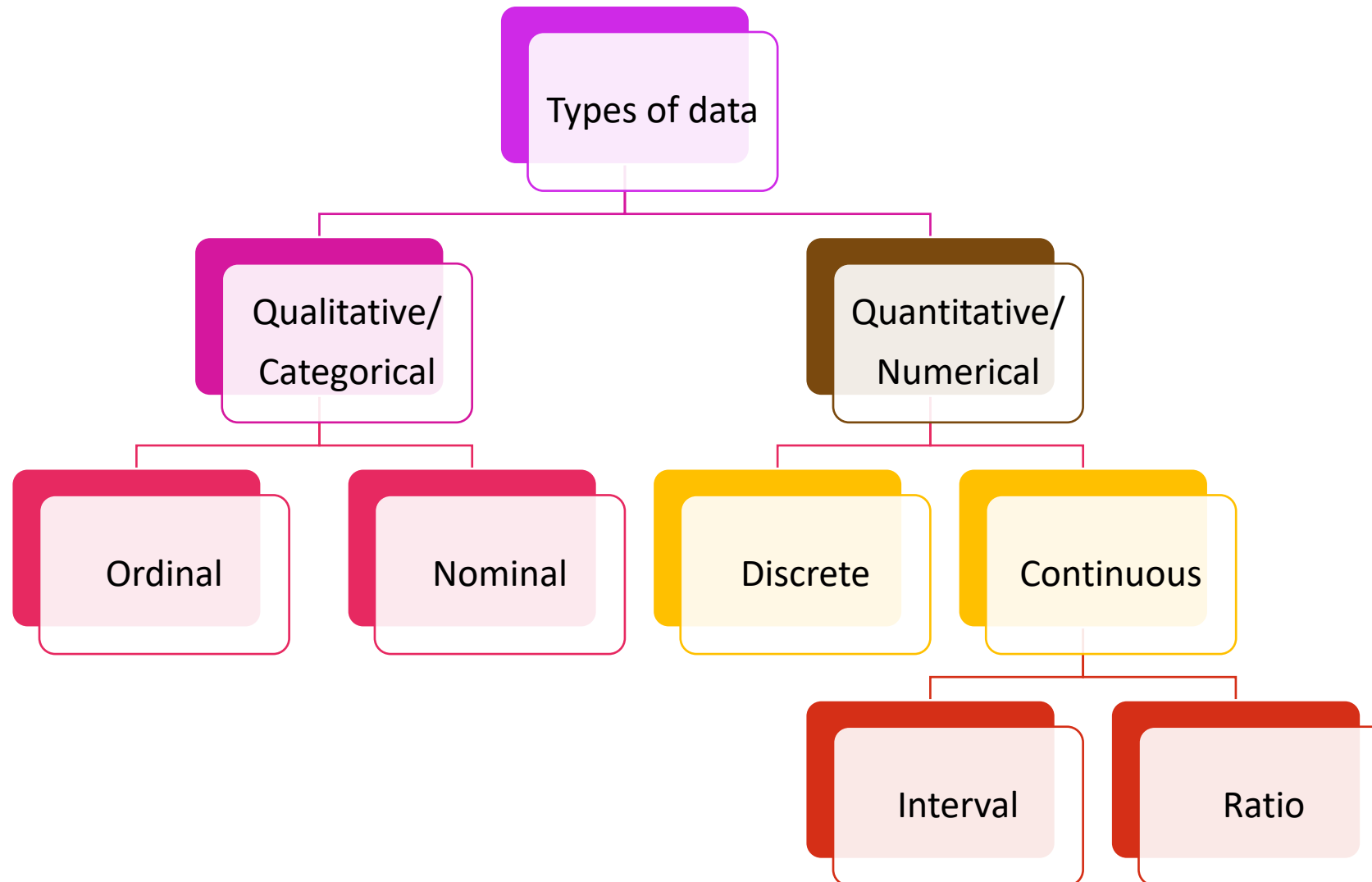
- Order placement
- Payment processing
- Order fulfillment
- Customer registration and authentication
- Promotion and discounts
- Returns and refund



Data

- Data refers to raw facts or values.
 - It is the raw material from which information and knowledge are derived.
 - Eg:
 - ✓ **Numeric Data:** Heights, weights, temperatures, etc.
 - ✓ **Categorical Data:** Gender, colors, types of fruits, etc.
 - ✓ **Text Data:** Articles, books, social media posts, etc.
 - ✓ **Image Data:** Photographs, medical images, satellite images, etc.
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Types of Data



Types of Data cont..

- **Quantitative (Numeric):** Data represented by numbers, such as measurements or counts.

E.g. count, height, weight, income, revenue, population, speed, etc.

- **Qualitative (Categorical):** Data represented by categories or labels, not in numerical form.

E.g. colour, gender, marital status, religion, job, education level, etc.



Types of Data cont..

- **Discrete data:** whole numbers and are often the result of counting.

E.g.

The number of currently working projects is 10

Number of bugs found during the testing is 30.

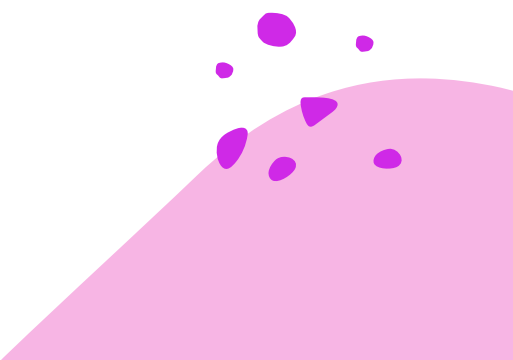
- **Continuous data:** can take on any value within a given range.

E.g.

Response time is 2.3 seconds.

Temperature: 23.5°C, 27.8°C, 28°C

Speed: 45.6 km/h or 55 mph, etc.



Types of Data cont..

- **Ordinal data:** Categorical/Qualitative data with a meaningful order.

E.g.

Customer ratings: 1 to 5 stars

Service Quality: Poor, Fair, Good, Very Good, Excellent

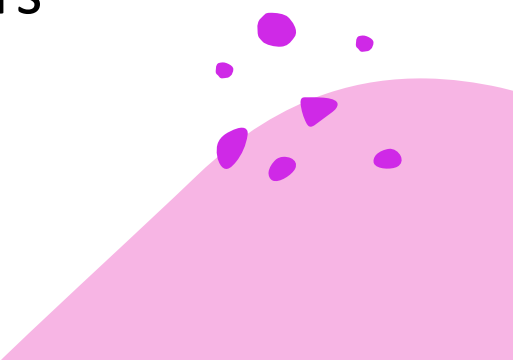
- **Nominal data:** Categorical/Qualitative data with no inherent order.

E.g.

Customer Data: Customer names, email addresses, phone numbers

Marital status: single, married, and divorced

Occupation: Teacher, Doctor, Plumber



Find out the types of data

1. Number of students in a class
2. Weight of a package
3. Types of fruits in a basket (e.g., apple, banana, orange)
4. Categories of books in a library (e.g., fiction, non-fiction, mystery)
5. Time taken to complete a race
6. Educational levels (e.g., high school, college, graduate)
7. Quantity of apples in a basket
8. Number of goals scored in a soccer match
9. Volume of water in a swimming pool
10. Gender of students in a class (e.g., male or female)

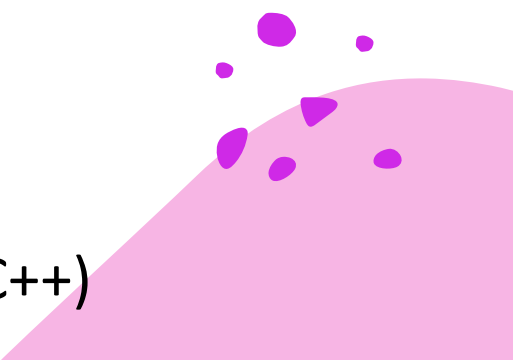


Activity

- Find out some data at BCAS Campus and categorize them as ordinal data, nominal data, discrete data and continuous data.



Find out the types of data

1. Number of employees attending a training session
 2. Cost of software licenses
 3. Types of software defects (e.g., UI bugs, performance issues)
 4. Level of customer satisfaction (e.g., satisfied, neutral, dissatisfied)
 5. Number of tickets in a helpdesk queue
 6. Duration of a system downtime (e.g., in minutes)
 7. Types of software development methodologies (e.g., Agile, Waterfall, Scrum)
 8. Number of software releases in a year
 9. Number of active users on a website
 10. Types of programming languages used in a project (e.g., Python, Java, C++)
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Information

- Information refers to data that has been processed, organized, or structured in a meaningful way to convey knowledge or communicate a message.

E.g

Source Code

System Requirement Specification(SRS)

Sales performance report

Health report

Financial report, etc.



Data vs Information

Data	Information
Data is raw facts	Processed from data
May or may not be meaningful	Always meaningful
Input to any system	Output after processing
Understanding is difficult	Understanding is easy
May not be in order	Should be in order

Example

- Date: 22021732
 - ✓ This has no meaning.
- Context:
 - ✓ It is 22nd February 1732
 - ✓ Still has no meaning. So, it's not an information.
- Meaning: 22ns February 1732 is the birthday of George Washington
 - ✓ This gives a meaning. So it's an information.



Structured and Unstructured data

Structured Data



Structured data is **quantitative** data in the form of numbers and values.

Unstructured Data



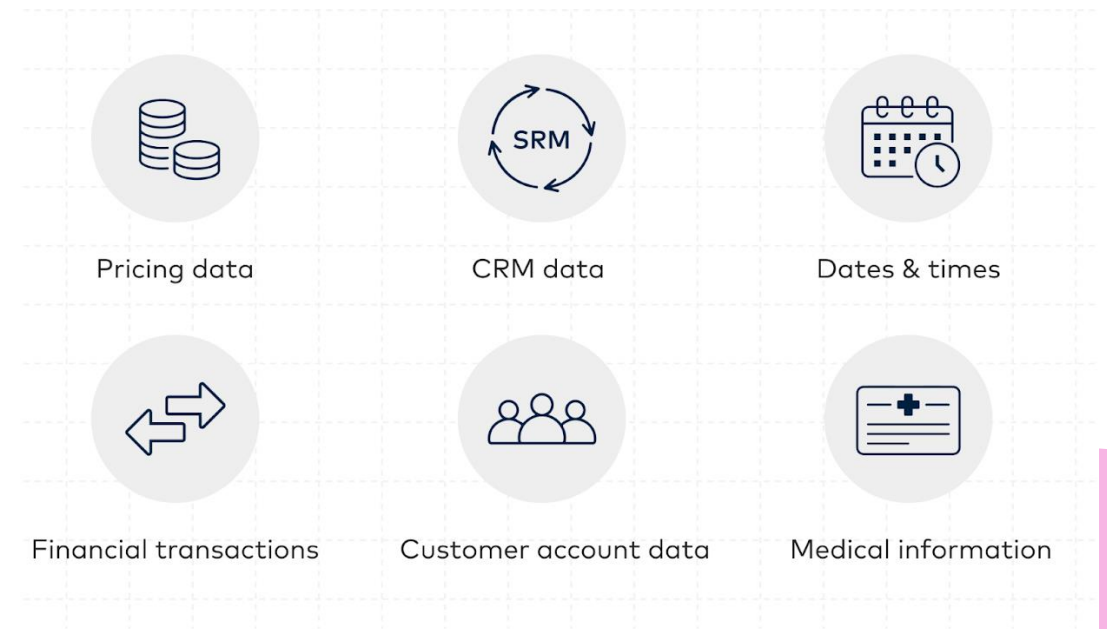
Unstructured data is **qualitative** data in the form of text files, audio files, video files.

Structured data

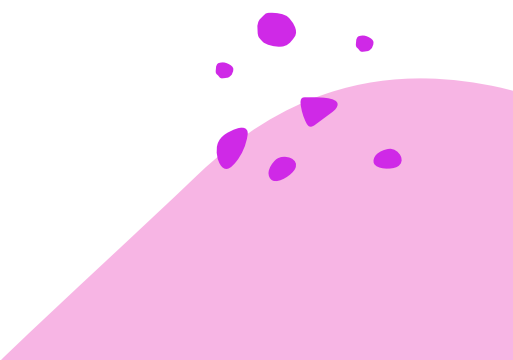
- Structured Data refers to any data that has a particular format.
- It is most often categorized as **quantitative** data, and it's the type of data most of us are used to working with.
- Eg:
 - ✓ Names, dates, addresses, credit card numbers, stock information, geolocation, and more.
 - ✓ Database entries, Spreadsheet entries and CSV entries

Structured data

- Structured data is highly organized and easily understood by **machine language**.
- Structured Query Language (SQL) is used to handle structured data.



Unstructured data (big data)

- Commonly, unstructured data refers to binary data.
 - It is often categorized as **qualitative** and cannot be processed and analyzed using conventional data tools and methods.
 - Eg:
 - ✓ Text documents
 - ✓ Images
 - ✓ Videos and audios files
 - ✓ Social media feeds
 - ✓ Sensor data
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Unstructured data

- Non-relational or NoSQL databases are the best fit for managing unstructured data.
- Unstructured data requires more advanced techniques such as
 - ✓ Natural language processing (NLP),
 - ✓ Image recognition, and
 - ✓ Machine learning algorithmsto extract insights and patterns



Structured Data

VS

Unstructured Data

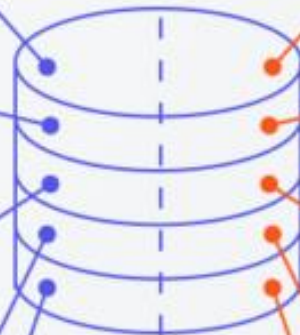
Can be displayed in rows, columns
and relational databases

Numbers, dates and strings

Estimated 20% of enterprise
data (Gartner)

Requires less storage

Easier to manage and protect
with legacy solutions



Cannot be displayed in rows,
columns and relational databases

Images, audio, video, word processing
files, e-mails, spreadsheets

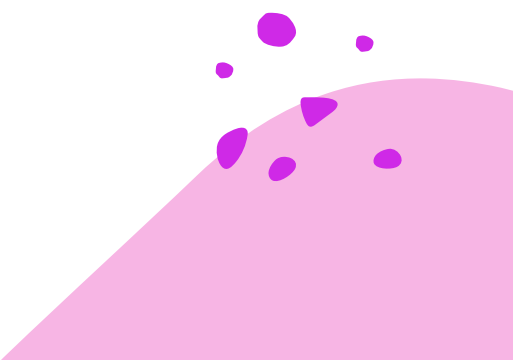
Estimated 80% of enterprise
data (Gartner)

Requires more storage

More difficult to manage and
protect with legacy solutions

Why data is so important?

Data in an organization add value to decision-making processes.

- Boosts productivity
 - Reduces the chances of data loss
 - Improves decision-making (strategic, tactical, and operational decisions)
 - To find solutions to problems
 - To back up your arguments
 - Helps for predictions
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Activity

Discuss what is

- ✓ Strategic
- ✓ Tactical
- ✓ Operational Decisions



Thank You!

